

Electronic Supplement: A Systematic Survey and Taxonomy of Prompt Engineering Evaluation Frameworks for Large Language Models

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About this supplement. This file contains Appendices A, B, and C of the main paper. Cross-references to sections and tables in the main paper are given as static text (e.g., “Section 7.5 of the main paper”) because this supplement compiles independently.

Appendices

A Evaluation Design Checklist for Prompt Engineering Studies

The following checklist is intended for use before finalizing an evaluation protocol for a prompt engineering study. Each group corresponds to one taxonomy dimension. Items are written as yes/no questions; a “no” answer does not disqualify a study but should be accompanied by explicit justification in the paper.

D1 - Evaluation Method

- Have you explicitly named the evaluation method your study uses (benchmark, human evaluation, LLM-as-judge, ablation study, or user study) in the methodology section?
- If you use benchmark evaluation, have you verified that the chosen benchmark’s task type and answer format match your prompting technique’s target capability?
- If you use LLM-as-judge, have you reported the judge model, version, prompt template, and scoring rubric in sufficient detail for replication?
- If you use human evaluation, have you reported annotator qualifications, inter-annotator agreement, and the number of evaluated items?

D2 - Task Scope

- Have you stated the task domain (e.g., reasoning, code generation, clinical QA, multilingual generation) explicitly, and confirmed it is reflected in your evaluation setup?
- If your technique targets a specialized domain (clinical, legal, multilingual), have you used or adapted evaluation instruments designed for that domain rather than borrowing general NLP metrics without justification?
- If you report results on multiple task types, have you analyzed performance separately per task type rather than reporting only an aggregate?

D3 - Metric Type

- Have you stated which metric type your study uses (reference-based, model-based, LLM-as-judge, or task-specific) and explained why it is appropriate for the task?

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- If you use reference-based metrics (BLEU, ROUGE, exact match), have you confirmed that a gold reference answer or label exists for every evaluated instance?
- If your task involves open-ended generation without a single correct answer, have you justified why your chosen metric adequately captures the relevant quality dimensions?
- Have you reported metric values with confidence intervals or significance tests where sample size permits?

D4 - Automation Level

- Have you characterized the automation level of your evaluation (fully manual, human-in-loop, hybrid, or fully automated)?
- If your evaluation is fully automated and uses a proprietary LLM as judge or scorer, have you documented the API version or snapshot date so that the exact judge state is recoverable?
- If any part of the evaluation involved human review (e.g., spot-checking automated scores), have you described the sampling procedure and what was reviewed?
- Have you considered whether a change in the judge model between submission and peer review could alter reported results?

D5 - Evaluation Scope

- Have you stated the evaluation scope explicitly (single-model/dataset, cross-model, cross-dataset, cross-domain, or cross-lingual)?
- If you report a technique as generally effective, have you evaluated it on at least two distinct models or two distinct datasets to support that generalizability claim?
- If results are reported on a single model, have you framed conclusions accordingly (e.g., “on GPT-4” rather than “across LLMs”)?
- If your study targets multilingual or cross-lingual evaluation, have you confirmed that your metric and judge model perform reliably in the target languages?

D6 - Robustness and Prompt Sensitivity

- Have you reported performance under at least one semantically equivalent paraphrase of the headline prompt, or explained why such reporting is not applicable?
- If your evaluation pipeline includes an LLM-as-judge component, have you reported the judge’s sensitivity to scoring-prompt rephrasing (e.g. JSS or equivalent) [8]?
- For headline numerical claims (“technique X improves accuracy by Y%”), have you confirmed that the claimed improvement survives the paraphrase variance, or framed the claim accordingly (“on this prompt phrasing”)?

D7 - Reproducibility and Contamination Resistance

- Have you released the prompts, exemplar sets, model configurations, and random seeds needed for a third party to re-run your evaluation?
- If any LLM-as-judge component is involved, have you reported the judge model identity, API version or snapshot date, and judge prompt template?
- Have you verified, by means stronger than a generic claim, that the test set used did not appear in any training corpus the underlying model has plausibly seen (e.g., dynamic benchmark construction [63], contamination probes, or held-out construction)?
- If a contamination check is not feasible, have you reported the limitation explicitly?

This checklist is available in machine-readable form at <https://github.com/rohithreddybc/PromptEvalTaxonomy>. Researchers may copy and adapt it for their own study preregistration or supplementary materials, with citation to this work.

B Representative Per-Paper Taxonomy Mapping (All Seven Dimensions)

Table B.1 presents a representative cross-section of the 152-paper corpus coded across all seven taxonomy dimensions. Ten papers are shown here – one per major paper category, selected to illustrate the range of configurations in the corpus. The complete 152-paper machine-readable dataset is included as a review-time snapshot with the submission; a citable tagged version will be released at acceptance to <https://github.com/rohithreddybc/PromptEvalTaxonomy> (see the Data Availability statement in the main paper). D6 (Robustness) and D7 (Reproducibility) codings were dual-coded with inter-rater $\kappa = 0.76$ and 0.78 respectively (see Section 7.5 of the main paper). Abbreviations: BM = Benchmark; LJ = LLM-as-judge; HE = Human eval; Abl = Ablation; TS = Task-specific; MB = Model-based; Hyb = Hybrid; Auto = Automated; HiL = Human-in-loop; CM = Cross-model; CD = Cross-domain; CDs = Cross-dataset; SM = Single-model; Sng = Single-prompt; MAg = Multi-prompt aggregate; SB = Sensitivity-bounded; AI = Artifact-incomplete; AR = Artifact-released; CC = Contamination-checked; JP = Judge-version-pinned; CC+JP = both properties satisfied.

Table B.1. Representative per-paper mapping to all seven taxonomy dimensions (D1–D7). Ten papers from the 152-paper corpus are shown, one per major category, selected to span the three archetypes (Section 8.9 of the main paper) and the five paper categories of Table 8 of the main paper. The complete 152-paper machine-readable dataset is included as a review-time snapshot with the submission; a citable tagged version will be released at acceptance to the GitHub URL above (see Data Availability statement in the main paper). D6 and D7 codings reflect the dual-coded inter-rater agreement of Section 7.5 of the main paper (κ = 0.76 and 0.78 respectively).

Paper (year)	D1 Method	D2 Task	D3 Metric	D4 Auto	D5 Scope	D6 Robust.	D7 Reprod.
<i>Prompting Technique Papers</i>							
Wei et al. [125] (2022) Chain- of-Thought	BM	Reasoning	TS	Auto	CM	Sng	AR
Kojima et al. [53] (2022) Zero-Shot CoT	BM	Reasoning	TS	Auto	CM	Sng	AR
Ouyang et al. [82] (2022) In-structGPT/RLHF	HE	Multi-task	Hyb	HiL	SM	Sng	AI
<i>Automated Prompt Optimization Papers</i>							
Zhou et al. [149] (2022) APE	BM	Multi-task	TS	Auto	CM	Sng	AR
<i>LLM-as-Judge / Metric Papers</i>							
Zheng et al. [145] (2023) MT-Bench	LJ	Multi-task	LJ	Auto	CM	MAG	JP
Liu et al. [70] (2023) G-Eval	LJ	Generation	LJ	Hyb	CM	Sng	JP+AR
<i>Benchmark / Dynamic Benchmark Papers</i>							
Lin et al. [63] (2024) Wild-Bench	LJ	Multi-task	LJ	Auto	CM	MAG	CC+AR
Wang et al. [118] (2018) GLUE	BM	Multi-task	TS	Auto	CDs	Sng	AR
<i>Robustness / Sensitivity Papers</i>							
Zhu et al. [151] (2023) PromptBench	BM	Multi-task	TS	Auto	CM	SB	AR

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Table B.1 (continued)

Paper (year)	D1 Method	D2 Task	D3 Metric	D4 Auto	D5 Scope	D6 Robust.	D7 Reprod.
LLM Evaluation Survey Papers							
Chang et al. [15] (2023) LLM HE Eval Survey		Multi-task	Hyb	HiL	CD	Sng	AI

C Comprehensive Per-Paper Mapping (152 Papers, All Seven Dimensions)

For full transparency, this appendix presents the complete corpus of 152 papers coded across all seven taxonomy dimensions (D1–D7). The representative ten-paper sample in Appendix B above (Table B.1) provides category-grouped expanded notes; the underlying machine-readable spreadsheet is included as a review-time snapshot with the submission; a citable tagged version will be released at acceptance to <https://github.com/rohithreddybc/PromptEvalTaxonomy> (see Data Availability statement in the main paper). Papers 80–99 were added in the first expansion round, 100–110 in the second, 111–145 in the third, and 146–152 in the fourth (Section 7 of the main paper, Table 4 therein), targeting LLM-judge quality, benchmark contamination, multi-prompt sensitivity, agentic, multilingual, clinical, prompt-injection, contamination-resistant, and CoT-faithfulness work underrepresented in the primary search results. D6 and D7 entries reflect the dual-coded inter-rater agreement of Section 7.5 of the main paper ($\kappa = 0.76$ and 0.78 respectively). Abbreviations follow those used in Appendix B above: D6 = Sng (Single-prompt), MAg (Multi-prompt aggregate), SB (Sensitivity-bounded), NR (Not reported); D7 = AR (Artifact-released), CC (Contamination-checked), JP (Judge-version-pinned), NR (Not reported); combined codes such as “CC+AR” or “JP+AR” indicate that both properties are satisfied.

Table C.1. Comprehensive per-paper mapping of the 152-paper corpus across all seven taxonomy dimensions. D1 (Evaluation Method), D2 (Task Scope), D3 (Metric Type), D4 (Automation Level), D5 (Evaluation Scope), D6 (Robustness), D7 (Reproducibility). D6 and D7 inter-rater $\kappa = 0.76$ and 0.78 respectively. Papers 1–79 were retained from the primary search round; papers 80–99 added in expansion round 1 (LLM-judge quality, contamination, multi-prompt sensitivity); 100–110 in round 2 (agentic, multilingual, prompt-injection, CoT-faithfulness); 111–145 in round 3 (foundation-model reports, RAG, safety, fairness, calibration, clinical, legal); and 146–152 in round 4 (long-context, tool-use, reward-model evaluation). See Table 4 of the main paper for the full expansion history.

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
<i>Core Prompting Technique Papers</i>									
1	Wei et al. [125]	Chain-of-Thought	Benchmark	Reasoning	Task-specific	Automated	Cross-model	Sng	AR
2	Kojima et al. [53]	Zero-Shot CoT	Benchmark	Reasoning	Task-specific	Automated	Cross-model	Sng	AR
3	Wang et al. [121]	Self-Consistency	Benchmark	Reasoning	Task-specific	Automated	Cross-model	Sng	AR
4	Yao et al. [134]	Tree of Thoughts	Benchmark	Reasoning	Task-specific	Automated	Single-model	Sng	AR
5	Zhou et al. [147]	Least-to-Most	Benchmark	Reasoning	Task-specific	Automated	Cross-model	Sng	AR
6	Besta et al. [9]	Graph of Thoughts	Benchmark	Reasoning	Task-specific	Automated	Single-model	Sng	AR
7	Wang et al. [119]	Plan-and-Solve	Benchmark	Reasoning	Task-specific	Automated	Single-model	Sng	AR
8	Lyu et al. [71]	Faithful CoT	Benchmark	Reasoning	Task-specific	Automated	Cross-model	Sng	AR
9	Besta et al. [10]	Chains/Trees/Graphs	Benchmark	Reasoning	Task-specific	Automated	Cross-model	NR	NR
10	Lewis et al. [58]	RAG	Benchmark	QA	Task-specific	Automated	Single-model	Sng	AR
11	Shin et al. [102]	AutoPrompt	Benchmark	QA, Reasoning	Task-specific	Automated	Single-model	Sng	AR
12	Agarwal et al. [3]	Many-Shot ICL	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
13	Ouyang et al. [82]	InstructGPT / RLHF	Human eval	Multi-task	Hybrid	Human-in-loop	Single-model	Sng	AI
14	Wei et al. [22]	FLAN	Benchmark	Multi-task	Task-specific	Automated	Single-model	Sng	AR
15	Li et al. [60]	Structured CoT	Benchmark	Code	Task-specific	Automated	Cross-model	Sng	AR
16	Ferrão et al. [32]	Manual Prompt Study	Benchmark	Code	Task-specific	Hybrid	Single-model	Sng	AR
17	Agrawal et al. [4]	GEPA	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
18	Agarwal et al. [2]	PromptWizard	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
19	White et al. [127]	Prompt Pattern Catalog	Ablation	Multi-task	Task-specific	Manual	Single-model	NR	NR
20	Dong et al. [26]	In-Context Learning	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
<i>Automated Prompt Optimization Papers</i>									
21	Zhou et al. [149]	APE	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR

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Table C.1 (continued)

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
22	Pryzant et al. [85]	APO	Benchmark	Multi-task	Task-specific	Automated	Single-model	Sng	AR
23	Deng et al. [25]	RLPrompt	Benchmark	Multi-task	Task-specific	Automated	Single-model	Sng	AR
24	Li et al. [61]	Auto Prompt Survey	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
25	Ramnath et al. [89]	Auto Prompt Opt. Survey	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
26	He et al. [42]	Context Engineering	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
27	Schulhoff et al. [98]	The Prompt Report	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
<i>Prompt Engineering Survey Papers</i>									
28	Sahoo et al. [93]	PE Techniques Survey	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
29	Vatsal & Dubey [114]	PE Methods Survey	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
30	Wen et al. [126]	PE Taxonomy	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
31	Sasaki et al. [97]	PE Patterns (SE)	Benchmark	Code	Task-specific	Automated	Cross-domain	NR	NR
32	Tian et al. [110]	Prompt Defect Taxonomy	Ablation	Code	Task-specific	Manual	Single-model	NR	NR
33	Chang et al. [15]	LLM Evaluation Survey	Human eval	Multi-task	Hybrid	Human-in-loop	Cross-domain	NR	NR
<i>LLM Evaluation Survey Papers</i>									
34	Guo et al. [40]	LLM Eval. Comprehensive	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
35	Ni et al. [80]	LLM Benchmark Survey	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
36	Mohammadi et al. [79]	LLM Agent Eval. Survey	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
37	Gururangan et al. [41]	Benchmark Adequacy	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
38	Polo et al. [150]	Benchmark ²	Benchmark	Multi-task	Task-specific	Automated	Cross-dataset	MAG	AR
39	Zhao et al. [143]	LLM Explainability	Human eval	Multi-task	Hybrid	Human-in-loop	Cross-domain	NR	NR

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Table C.1 (continued)

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
40	Wang et al. [118]	GLUE	Benchmark	Multi-task	Task-specific	Automated	Cross-dataset	Sng	AR
<i>Benchmark Papers</i>									
41	Wang et al. [117]	SuperGLUE	Benchmark	Multi-task	Task-specific	Automated	Cross-dataset	Sng	AR
42	Srivastava et al. [105]	BIG-Bench	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAg	AR
43	Hendrycks et al. [43]	MMLU	Benchmark	Multi-task	Task-specific	Automated	Cross-dataset	Sng	AR
44	Suzgun et al. [108]	BIG-Bench Hard	Benchmark	Reasoning	Task-specific	Automated	Cross-dataset	Sng	AR
45	Zhu et al. [151]	PromptBench	Benchmark	Multi-task	Task-specific	Automated	Cross-model	SB	AR
46	Wang et al. [122]	MMLU-Pro	Benchmark	Multi-task	Task-specific	Automated	Cross-dataset	Sng	AR
47	Lin et al. [63]	WildBench	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	MAg	CC+AR
48	Gao et al. [35]	RAG Survey	Benchmark	QA	Task-specific	Automated	Cross-domain	NR	NR
49	Zheng et al. [145]	MT-Bench / Chatbot Arena	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	MAg	JP+AR
<i>LLM-as-Judge Papers</i>									
50	Dubois et al. [27]	AlpacaEval 2.0	LLM-as-Judge	Generation	LLM-judge	Automated	Cross-model	Sng	JP+AR
51	Liu et al. [70]	G-Eval	LLM-as-Judge	Generation	LLM-judge	Hybrid	Cross-model	Sng	JP+AR
52	Gu et al. [38]	LLM-as-Judge Survey	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-domain	NR	NR
53	Li et al. [59]	LLM Judge Survey	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-domain	NR	NR
54	Feng et al. [31]	SAGE	LLM-as-Judge	Multi-task	LLM-judge	Hybrid	Cross-model	MAg	JP+AR
55	Tan et al. [109]	JudgeBench	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	MAg	JP+AR
56	Ye et al. [7]	LLMs vs. Humans Study	LLM-as-Judge, Human	Multi-task	LLM-judge	Hybrid	Cross-model	Sng	AR
57	Calderon et al. [13]	Alternative Annotator	LLM-as-Judge, Human	Multi-task	LLM-judge	Hybrid	Cross-model	MAg	AR
58	Chen et al. [17]	MAJ-EVAL	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-domain	MAg	AR
59	Yang et al. [131]	LLM Judge Empirical	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	MAg	AR
60	Ye et al. [135]	Judgement Bias Study	LLM-as-Judge, Human	Multi-task	LLM-judge	Hybrid	Cross-model	Sng	AR
61	Panickssery et al. [83]	Judging the Judges	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	Sng	AR
62	Chiang et al. [21]	Can LLMs Replace Humans	LLM-as-Judge, Human	Generation	LLM-judge	Hybrid	Single-model	Sng	AR

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Table C.1 (continued)

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
63	Min et al. [76]	FactScore	LLM-as-Judge	Generation	LLM-judge	Hybrid	Cross-model	Sng	AR
<i>Metric and Evaluation Quality Papers</i>									
64	Xu et al. [130]	InstructScore	LLM-as-Judge	Generation	LLM-judge	Automated	Cross-model	Sng	AR
65	Es et al. [28]	RAGAs	Benchmark	QA	Task-specific	Automated	Single-model	Sng	AR
66	Gehrmann et al. [36]	NLG Eval. Survey	Human eval	Generation	Model-based	Human-in-loop	Cross-domain	NR	NR
67	Zhong et al. [146]	Multi-Dim. Evaluator	Human eval	Generation	Hybrid	Hybrid	Cross-model	Sng	AR
68	Yu et al. [139]	DeCE	LLM-as-Judge	QA	LLM-judge	Automated	Cross-model	Sng	AR
69	Farquhar et al. [55]	Semantic Entropy	Benchmark	QA	Task-specific	Automated	Cross-model	Sng	AR
70	Gao et al. [87]	NLG Eval Status	LLM-as-Judge	Generation	LLM-judge	Automated	Cross-domain	NR	NR
71	Ram et al. [88]	In-Context RAG	Benchmark	QA	Task-specific	Automated	Cross-model	Sng	AR
72	Sclar et al. [99]	Prompt Format Sensitivity	Benchmark	Multi-task	Task-specific	Automated	Cross-model	SB	AR
<i>Prompt Sensitivity and Format Papers</i>									
73	Razavi et al. [90]	PromptSET	Benchmark	QA	Task-specific	Automated	Cross-model	SB	AR
74	Salinas & Gama [96]	Prompt Formatting Study	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR
75	Salinas & Morstatter [95]	Butterfly Effect Study	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR
76	Chen et al. [18]	Prompt Alchemy	Benchmark	Code	Task-specific	Automated	Cross-model	Sng	AR
77	He et al. [111]	Secure Code Prompting	Benchmark	Code	Task-specific	Hybrid	Cross-model	Sng	AR
<i>Code and Software Engineering Papers</i>									
78	Hou et al. [44]	LLMs for SE Survey	Benchmark	Code	Task-specific	Automated	Cross-model	NR	NR
79	Khojah et al. [51]	CodePromptEval	Benchmark	Code	Task-specific	Hybrid	Single-model	Sng	AR
<i>LLM-as-Judge Bias and Reliability Studies</i>									
80	Koo et al. [54]	Cognitive Biases in LLM Judges	LLM-judge	Multi-task	LLM-judge	Automated	Cross-model	MAG	AR
81	Ye et al. [136]	Justice or Prejudice	LLM-judge	Multi-task	LLM-judge	Automated	Cross-model	MAG	AR
82	Zhu et al. [152]	JudgeLM	LLM-judge	Multi-task	LLM-judge	Automated	Cross-model	MAG	AR
83	Verga et al. [116]	Panel-of-Judges Evaluation	LLM-judge	Multi-task	LLM-judge	Automated	Cross-model	MAG	AR
84	Wataoka et al. [123]	Self-Preference Bias in LLM-as-Judge	LLM-judge	Multi-task	LLM-judge	Automated	Cross-model	Sng	NR
<i>Benchmark Contamination and Dynamic Benchmarking Papers</i>									

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Table C.1 (continued)

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
85	Sainz et al. [94]	NLP Evaluation in Trouble	Benchmark	Multi-task	Task-specific	Automated	Cross-dataset	NR	CC
86	Deng et al. [24]	Investigating Data Contamination	Benchmark	Multi-task	Task-specific	Automated	Cross-dataset	NR	CC
87	Yang et al. [132]	Rethinking Benchmark Contamination	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAg	CC
88	Wang et al. [120]	Benchmark Self-Evolving	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	CC+AR
89	McIntosh et al. [75]	LLM Benchmark Inadequacies	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
90	Zeng et al. [142]	LLMBar (Eval. Instruction Following)	LLM-judge	Multi-task	LLM-judge	Automated	Cross-model	Sng	AR
91	Chern et al. [20]	ScaleEval (Agent Debate)	LLM-judge	Multi-task	LLM-judge	Automated	Cross-model	MAg	AR
92	Li et al. [62]	LLM-based NLG Evaluation Survey	Benchmark	Generation	Model-based	Automated	Cross-model	NR	NR
93	Zhou et al. [148]	Instruction-Following Eval (IFEval)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
94	Qin et al. [86]	ChatGPT Sentiment Reasoning	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
<i>Multi-Prompt, Clinical, and Multimodal Evaluation Papers</i>									
95	Mizrahi et al. [78]	State of What Art (TACL)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAg	AR
96	Sivarajkumar et al. [103]	Clinical Prompting Evaluation (JMIR)	Benchmark	Clinical	Task-specific	Automated	Cross-model	MAg	NR
97	Polo et al. [84]	PromptEval (Multi-Prompt Eval)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	SB	AR
98	Jang & Lukasiewicz [49]	Consistency Analysis of ChatGPT	Human eval	Multi-task	Hybrid	Manual	Cross-domain	NR	NR
99	Fu et al. [33]	MME Multimodal Evaluation	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
<i>Judge Sensitivity and Multilingual Evaluation Papers</i>									
100	Bellibaltu et al. [8]	JudgeSense (Judge Sensitivity)	Benchmark	Multi-task	LLM-judge	Automated	Cross-model	SB	AR
101	Vatsal & Singh [115]	Multilingual PE Survey	Benchmark	Multilingual	Task-specific	Automated	Cross-domain	NR	NR
<i>Agentic, Tool-Use, and Safety Evaluation Papers</i>									
102	Liu et al. [68]	AgentBench	Benchmark	Agentic	Task-specific	Automated	Cross-model	Sng	AR
103	Ma et al. [72]	AgentBoard	Benchmark	Agentic	Task-specific	Automated	Cross-model	MAg	AR
104	Liu et al. [69]	HouYi (Prompt Injection)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAg	AR

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Table C.1 (continued)

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
105	Maloyan et al. [74]	Judge Vulnerability to Injection	LLM-judge	Multi-task	LLM-judge	Automated	Cross-model	MAg	AR
<i>Chain-of-Thought Faithfulness and Hallucination Studies</i>									
106	Lanham et al. [57]	CoT Faithfulness Measurement	Benchmark	Reasoning	Task-specific	Automated	Cross-model	MAg	AR
107	Chen et al. [19]	Reasoning Models Faithfulness	Benchmark	Reasoning	Task-specific	Automated	Cross-model	SB	AR
108	Akbar et al. [5]	HalluMeasure	LLM-judge	Generation	LLM-judge	Automated	Cross-model	Sng	AR
109	Liu et al. [65]	RH-Bench (Multimodal Hallu.)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAg	AR
<i>Multilingual LLM-as-Judge Meta-Evaluation</i>									
110	Son et al. [104]	MM-Eval (Multilingual Judge)	LLM-judge	Multilingual	LLM-judge	Automated	Cross-model	MAg	AR
<i>Foundation-Model Technical Reports</i>									
111	Achiam et al. [1]	GPT-4 Technical Report	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	Sng	NR
112	Touvron et al. [112]	LLaMA Foundation Models	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
113	Touvron et al. [113]	Llama 2 Open Foundation Models	Benchmark	Multi-task	Hybrid	Hybrid	Cross-model	MAg	AR
114	Hurst et al. [47]	GPT-4o System Card	Benchmark	Multi-task	Hybrid	Automated	Cross-domain	Sng	NR
115	Bubeck et al. [12]	Sparks of AGI (GPT-4 study)	Human eval	Multi-task	Hybrid	Manual	Cross-domain	Sng	NR
<i>Mathematical Reasoning Benchmarks</i>									
116	Cobbe et al. [23]	GSM8K Grade-School Math	Benchmark	Reasoning	Task-specific	Automated	Cross-model	Sng	AR
117	Lin et al. [64]	TruthfulQA	Benchmark	Multi-task	Task-specific	Hybrid	Cross-model	Sng	AR
118	Mirzadeh et al. [77]	GSM-Symbolic	Benchmark	Reasoning	Task-specific	Automated	Cross-model	SB	AR
119	Shi et al. [101]	MGSM (Multilingual CoT)	Benchmark	Multilingual	Task-specific	Automated	Cross-model	Sng	AR
<i>Code Generation Evaluation Papers</i>									
120	Jimenez et al. [50]	SWE-bench	Benchmark	Code	Task-specific	Automated	Cross-model	Sng	AR
121	Liu et al. [66]	EvalPlus / HumanEval+	Benchmark	Code	Task-specific	Automated	Cross-model	MAg	AR
122	Jain et al. [48]	LiveCodeBench	Benchmark	Code	Task-specific	Automated	Cross-model	Sng	CC+AR

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Table C.1 (continued)

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
123	Rozière et al. [91]	Code Llama	Benchmark	Code	Task-specific	Automated	Cross-model	Sng	AR
Retrieval-Augmented Generation Evaluation Papers									
124	Saad-Falcon et al. [92]	ARES	LLM-judge	QA	LLM-judge	Automated	Cross-model	MAG	AR
125	Yu et al. [140]	Auepora RAG Survey	Benchmark	Multi-task	Hybrid	Automated	Cross-domain	NR	NR
126	Chen et al. [16]	RGB (RAG Benchmark)	Benchmark	QA	Task-specific	Automated	Cross-model	MAG	AR
127	Xiong et al. [128]	MIRAGE / MedRAG (Medical)	Benchmark	Clinical	Task-specific	Automated	Cross-model	MAG	AR
128	Zhao et al. [144]	RAG-for-AIGC Survey	Benchmark	Multi-task	Hybrid	Automated	Cross-domain	NR	NR
Safety, Red-Teaming, and Jailbreak Evaluation Papers									
129	Wei et al. [124]	Jailbroken (Safety Failure Modes)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR
130	Yu et al. [141]	GPTFuzzer	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR
131	Bhardwaj & Poria [11]	RED-EVAL (Chain of Utterances)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR
132	Huang et al. [46]	Catastrophic Jailbreak	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR
133	Sharma et al. [100]	Constitutional Classifiers	Benchmark	Multi-task	Task-specific	Automated	Cross-model	SB	AR
134	Yi et al. [138]	Jailbreak Attacks & Defenses Survey	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
Bias and Fairness Evaluation Papers									
135	Gallegos et al. [34]	Bias & Fairness in LLMs Survey	Benchmark	Multi-task	Hybrid	Hybrid	Cross-domain	NR	NR
136	Esiobu et al. [29]	ROBBIE (Bias Evaluation)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR
Calibration and Uncertainty Quantification Papers									
137	Xiong et al. [129]	LLM Confidence Elicitation	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR
138	Geng et al. [37]	Confidence Calibration Survey	Benchmark	Multi-task	Hybrid	Automated	Cross-domain	NR	NR
139	Steyvers et al. [106]	Calibration Gap (Nature MI)	Human eval	Multi-task	Hybrid	Hybrid	Cross-model	MAG	NR
Legal Domain Evaluation Papers									
140	Guha et al. [39]	LegalBench	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR

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Table C.1 (continued)

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
141	Fei et al. [30]	LawBench (Chinese Legal)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
Medical and Cognitive Evaluation Papers									
142	Nori et al. [81]	GPT-4 on Medical Challenges	Benchmark	Clinical	Task-specific	Hybrid	Single-model	MAg	NR
143	Strachan et al. [107]	Theory of Mind (Nature HB)	Human eval	Multi-task	Hybrid	Hybrid	Cross-model	MAg	AR
144	Chang et al. [14]	Red Teaming ChatGPT in Medicine	Human eval	Clinical	Hybrid	Manual	Cross-model	MAg	NR
145	Kim et al. [52]	Medical Hallucinations Eval	Benchmark	Clinical	Task-specific	Hybrid	Cross-model	MAg	AR
Long-Context Evaluation Papers									
146	Hsieh et al. [45]	RULER (Real Context Size)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	SB	AR
147	Bai et al. [6]	LongBench (Bilingual)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
148	Yen et al. [137]	HELMET (Long-Context Framework)	Benchmark	Multi-task	Hybrid	Hybrid	Cross-model	MAg	AR
Tool-Use and Function-Calling Evaluation Papers									
149	Yao et al. [133]	τ -bench (Tool-Agent-User)	Benchmark	Agentic	Task-specific	Automated	Cross-model	MAg	AR
150	Liu et al. [67]	ToolACE (Function Calling)	Benchmark	Agentic	Task-specific	Automated	Cross-model	MAg	AR
Reward Model and Preference Evaluation Papers									
151	Lambert et al. [56]	RewardBench	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
152	Malik et al. [73]	RewardBench 2	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAg	AR

References

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altmenschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. GPT-4 Technical Report. [arXiv:2303.08774](#) [cs.CL]
- [2] Eshaan Agarwal et al. 2024. PromptWizard: Task-Aware Prompt Optimization Framework. *arXiv preprint arXiv:2405.18369* (2024).
- [3] Rishabh Agarwal, Avi Singh, Lei M. Zhang, Bernd Bohnet, Luis Rosias, Stephanie Chan, Biao Zhang, Ankesh Anand, Zaheer Abbas, Azade Nova, et al. 2024. Many-Shot In-Context Learning. *arXiv preprint arXiv:2404.11018* (2024).
- [4] Lakshya A. Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziem, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J. Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. 2025. GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning. *arXiv preprint arXiv:2507.19457* (2025). Accepted at ICLR 2026 (Oral).
- [5] Shayan A. Akbar, Mahmoud Ghaffarzadegan, and Maryam Habibi. 2024. HalluMeasure: Fine-Grained Hallucination Measurement Using Chain-of-Thought Reasoning. [arXiv:2410.04339](#) [cs.CL]
- [6] Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. 2023. LongBench: A Bilingual, Multitask Benchmark for Long Context Understanding. *arXiv preprint arXiv:2308.14508* (2023).
- [7] Anna Bavaresco, Raffaella Bernardi, Leonardo Bertolazzi, Desmond Elliott, Raquel Fernández, Albert Gatt, Esam Ghaleb, Mario Giulianelli, Michael Hanna, Alexander Koller, et al. 2024. LLMs instead of Human Judges? A Large Scale Empirical Study across 20 NLP Evaluation Tasks. *arXiv preprint arXiv:2406.18403* (2024).
- [8] Rohith Reddy Bellibaltu, Edward Raff, and Wenbin Zhang. 2026. JudgeSense: A Benchmark for Prompt Sensitivity in LLM-as-a-Judge Systems. *arXiv preprint arXiv:2604.23478v2* (May 2026). [doi:10.48550/arXiv.2604.23478](#)
- [9] Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewiadomski, Piotr Nyczyk, and Torsten Hoeffler. 2023. Graph of Thoughts: Solving Elaborate Problems with Large Language Models. *arXiv preprint arXiv:2308.09687* (2023).
- [10] Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewiadomski, Piotr Nyczyk, and Torsten Hoeffler. 2024. Demystifying Chains, Trees, and Graphs of Thoughts. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2024). [arXiv:2401.14295](#).
- [11] Rishabh Bhardwaj and Soujanya Poria. 2023. Red-Teaming Large Language Models using Chain of Utterances for Safety-Alignment. *arXiv preprint arXiv:2308.09662* (2023).
- [12] Sébastien Bubeck, Varun Chandrasekaran, Ronen Eldan, Johannes Gehrke, Eric Horvitz, Ece Kamar, Peter Lee, Yin Tat Lee, Yuanzhi Li, Scott Lundberg, Harsha Nori, Hamid Palangi, Marco Tulio Ribeiro, and Yi Zhang. 2023. Sparks of Artificial General Intelligence: Early Experiments with GPT-4. *arXiv preprint arXiv:2303.12712* (2023).
- [13] Nitay Calderon, Roi Reichart, and Rotem Dror. 2025. The Alternative Annotator Test for LLM-as-a-Judge: How to Statistically Justify Replacing Human Annotators with LLMs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 16051–16081.
- [14] Crystal T. Chang, Hodan Farah, Haiwen Gui, Shiva Justin Rezaei, Charbel Bou-Khalil, Yu-Jiu Park, Akshay Raj, Saakshi Pope, et al. 2025. Red Teaming ChatGPT in Medicine to Yield Real-World Insights on Model Behavior. *NPJ Digital Medicine* (2025). [doi:10.1038/s41746-025-01486-5](#)
- [15] Yupeng Chang, Xu Wang, Jindong Wang, Yuan Wu, Linyi Yang, Kaijie Zhu, Hao Chen, Xiaoyuan Yi, Cunxiang Wang, Yidong Wang, et al. 2024. A Survey on Evaluation of Large Language Models. *ACM Transactions on Intelligent Systems and Technology* 15, 3 (2024). [arXiv:2307.03109](#).
- [16] Jiawei Chen, Hongyu Lin, Xianpei Han, and Le Sun. 2023. Benchmarking Large Language Models in Retrieval-Augmented Generation. *arXiv preprint arXiv:2309.01431* (2023).
- [17] Jiaju Chen, Yuxuan Lu, Xiaojie Wang, Huimin Zeng, Jing Huang, Jiri Gesi, Ying Xu, Bingsheng Yao, and Dakuo Wang. 2025. Multi-Agent-as-Judge: Aligning LLM-Agent-Based Automated Evaluation with Multi-Dimensional Human Evaluation. *arXiv preprint arXiv:2507.21028* (2025).
- [18] Yuqun Chen et al. 2025. Prompt Alchemy: Automatic Prompt Refinement for Enhancing Code Generation. *IEEE Transactions on Software Engineering* (2025). [arXiv:2412.09935](#).
- [19] Yanda Chen, Joe Benton, Ansh Radhakrishnan, Jonathan Uesato, Carson Denison, John Schulman, Arushi Somani, Peter Hase, Misha Wagner, Fabien Roger, Vlad Mikulik, Samuel R. Bowman, Jan Leike, Jared Kaplan, and Ethan Perez. 2025. Reasoning Models Don’t Always Say What They Think. [arXiv:2505.05410](#) [cs.CL]
- [20] Steffi Chern, Ethan Chern, and Pengfei Liu. 2024. Can Large Language Models be Trusted for Evaluation? Scalable Meta-Evaluation of LLMs as Evaluators via Agent Debate. [arXiv:2401.16788](#) [cs.CL]
- [21] Cheng-Han Chiang and Hung-yi Lee. 2023. Can Large Language Models Be an Alternative to Human Evaluations? *arXiv preprint arXiv:2305.01937* (2023).

- [22] Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Yunxuan Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, et al. 2022. Scaling Instruction-Finetuned Language Models. *arXiv preprint arXiv:2210.11416* (2022).
- [23] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training Verifiers to Solve Math Word Problems. *arXiv preprint arXiv:2110.14168* (2021).
- [24] Chunyuan Deng, Yilun Zhao, Xiangru Tang, Mark Gerber, Daniel Khashabi, Dian Yu, Bailin Chen, Qian Li, Rheeey Up-paal Wang, Shu Yu, et al. 2023. Investigating Data Contamination in Modern Benchmarks for Large Language Models. *arXiv:2311.09783* [cs.CL]
- [25] Mingkai Deng, Jianyu Wang, Cheng-Ping Hsieh, Yihan Wang, Han Gao, Terrance Ye, Meng Zhang, Rudrasis Bhatt, and Zhiting Mei. 2022. RLPrompt: Optimizing Discrete Text Prompts with Reinforcement Learning. *arXiv preprint arXiv:2205.12548* (2022).
- [26] Qingxiu Dong, Lei Li, Damai Dai, Ce Zheng, Zhiyong Wu, Baobao Chang, Xu Sun, Jingjing Xu, and Zhifang Sui. 2022. A Survey on In-context Learning. *arXiv preprint arXiv:2301.00234* (2022).
- [27] Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B. Hashimoto. 2024. Length-Controlled AlpacaEval: A Simple Way to Debias Automatic Evaluators. In *Proceedings of the Conference on Language Modeling (COLM)*. *arXiv:2404.04475*.
- [28] Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. 2023. RAGAs: Automated Evaluation of Retrieval Augmented Generation. *arXiv preprint arXiv:2309.15217* (2023).
- [29] David Esiobu, Xiaoqing Tan, Saghar Hosseini, Megan Ung, Yuchen Zhang, Jude Fernandes, Jane Dwivedi-Yu, Eleonora Presani, Adina Williams, and Eric Michael Smith. 2023. ROBBI: Robust Bias Evaluation of Large Generative Language Models. *arXiv preprint arXiv:2311.18140* (2023).
- [30] Zhiwei Fei, Xiaoyu Shen, Dawei Zhu, Fengzhe Zhou, Zhuo Han, Songyang Zhang, Kai Chen, Zongwen Shen, and Jidong Ge. 2023. LawBench: Benchmarking Legal Knowledge of Large Language Models. *arXiv preprint arXiv:2309.16289* (2023).
- [31] Yuanning Feng, Dongping Chen, et al. 2025. Are We on the Right Way to Assessing LLM-as-a-Judge? *arXiv preprint arXiv:2512.16041* (2025).
- [32] Rodrigo Ferrão et al. 2025. Manual Prompt Engineering is Not Dead: A Case Study on Large Language Models for Code Vulnerability Detection with DSPy. *2025 8th International Conference on Data Science and Machine Learning Applications (CDMA)* (2025).
- [33] Chaoyou Fu, Peixian Chen, Yunhang Shen, Yulei Qin, Mengdan Zhang, Xu Lin, Jinrui Yang, Xiwu Zheng, Ke Li, Xing Sun, Yunsheng Wu, and Rongrong Ji. 2023. MME: A Comprehensive Evaluation Benchmark for Multimodal Large Language Models. *arXiv:2306.13394* [cs.CV]
- [34] Isabel O. Gallegos, Ryan A. Rossi, Joe Barrow, Md Mehrab Tanjim, Sungchul Kim, Franck Dernoncourt, Tong Yu, Ruiyi Zhang, and Nesreen K. Ahmed. 2024. Bias and Fairness in Large Language Models: A Survey. *Computational Linguistics* (2024). *arXiv:2309.00770*.
- [35] Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. 2023. Retrieval-Augmented Generation for Large Language Models: A Survey. *arXiv preprint arXiv:2312.10997* (2023).
- [36] Sebastian Gehrmann, Elizabeth Clark, and Thibault Sellam. 2022. Repairing the Cracked Foundation: A Survey of Obstacles in Evaluation Practices for Generated Text. *arXiv preprint arXiv:2202.06935* (2022).
- [37] Jiahui Geng, Fengyu Cai, Yuxia Wang, Heinz Koeppl, Preslav Nakov, and Iryna Gurevych. 2023. A Survey of Confidence Estimation and Calibration in Large Language Models. *arXiv preprint arXiv:2311.08298* (2023).
- [38] Jiawei Gu, Xuhui Jiang, Zhichao Shi, Hexiang Tan, Xuehao Zhai, Chengjin Xu, Wei Li, et al. 2025. A Survey on LLM-as-a-Judge. *arXiv preprint arXiv:2411.15594* (2025). v6, revised October 2025.
- [39] Neel Guha, Julian Nyarko, Daniel E. Ho, Christopher Ré, Adam Chilton, Aditya Narayana, Alex Chohlas-Wood, Austin Peters, Brandon Waldon, Daniel N. Rockmore, Diego Zambrano, Dmitry Talisman, Enam Hoque, Faiz Surani, Frank Fagan, Galit Sarfaty, Gregory M. Dickinson, Haggai Porat, Jason Hegland, et al. 2023. LegalBench: A Collaboratively Built Benchmark for Measuring Legal Reasoning in Large Language Models. *arXiv preprint arXiv:2308.11462* (2023).
- [40] Zishan Guo, Renren Jin, Chuang Liu, Yufei Huang, Dan Shi, Linhao Yu, Yan Liu, Jiaxuan Li, Pengfei Xiong, Deyi Xiong, et al. 2023. Evaluating Large Language Models: A Comprehensive Survey. *arXiv preprint arXiv:2310.19736* (2023).
- [41] Suchin Gururangan et al. 2024. Inadequacies of Large Language Model Benchmarks in the Era of Generative Artificial Intelligence. *arXiv preprint arXiv:2402.09248* (2024).
- [42] Yifu He et al. 2025. A Survey of Context Engineering for Large Language Models. *arXiv preprint arXiv:2507.13334* (2025).
- [43] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring Massive Multitask Language Understanding. *arXiv preprint arXiv:2009.03300* (2021).

- [44] Xinyi Hou, Yanjie Zhao, Yue Liu, Zhou Yang, Kailong Wang, Li Li, Xiapu Luo, David Lo, John Grundy, and Haoyu Wang. 2024. Do Advanced Language Models Eliminate the Need for Prompt Engineering in Software Engineering? *ACM Transactions on Software Engineering and Methodology* (2024). arXiv:2411.02093.
- [45] Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekeshe, Fei Jia, Yang Zhang, and Boris Ginsburg. 2024. RULER: What's the Real Context Size of Your Long-Context Language Models? arXiv:2404.06654 [cs.CL]
- [46] Yangsibo Huang, Samyak Gupta, Mengzhou Xia, Kai Li, and Danqi Chen. 2023. Catastrophic Jailbreak of Open-source LLMs via Exploiting Generation. *arXiv preprint arXiv:2310.06987* (2023).
- [47] Aaron Hurst, Adam Lerer, Adam P. Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. 2024. GPT-4o System Card. arXiv:2410.21276 [cs.CL]
- [48] Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. 2024. LiveCodeBench: Holistic and Contamination Free Evaluation of Large Language Models for Code. arXiv:2403.07974 [cs.CL]
- [49] Myeongjun Jang and Thomas Lukasiewicz. 2023. Consistency Analysis of ChatGPT. *arXiv preprint arXiv:2303.06273* (2023).
- [50] Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. 2023. SWE-bench: Can Language Models Resolve Real-World GitHub Issues? arXiv:2310.06770 [cs.CL]
- [51] Ranim Khojah, Francisco Gomes de Oliveira Neto, Mazen Mohamad, and Philipp Leitner. 2025. The Impact of Prompt Programming on Function-Level Code Generation. *IEEE Transactions on Software Engineering* (2025). doi:10.1109/TSE.2025.3587794 arXiv:2412.20545.
- [52] Yubin Kim, Hyewon Jeong, Shan Chen, Shuyue Stella Li, Mingyu Lu, Kumail Alhamoud, Jimin Mun, Cristina Grau, Minseok Jung, Rodrigo R. Gameiro, et al. 2025. Medical Hallucinations in Foundation Models and Their Impact on Healthcare. *arXiv preprint arXiv:2503.05777* (2025).
- [53] Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large Language Models are Zero-Shot Reasoners. *arXiv preprint arXiv:2205.11916* (2022).
- [54] Ryan Koo, Minhwa Lee, Vipul Raheja, Jong Inn Park, Zae Myung Kim, and Dongyeop Kang. 2023. Benchmarking Cognitive Biases in Large Language Models as Evaluators. arXiv:2309.17012 [cs.CL]
- [55] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2024. Detecting Hallucinations in Large Language Models Using Semantic Entropy. *Nature* 630 (2024), 625–630.
- [56] Nathan Lambert, Valentina Pyatkin, Jacob Morrison, LJ Miranda, Bill Yuchen Lin, Khyathi Chandu, Nouha Dziri, Sachin Kumar, Tom Zick, Yejin Choi, Noah A. Smith, and Hannaneh Hajishirzi. 2024. RewardBench: Evaluating Reward Models for Language Modeling. *arXiv preprint arXiv:2403.13787* (2024).
- [57] Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Jackson Kernion, Kamile Lukosiute, Karina Nguyen, Newton Cheng, Nicholas Joseph, Nicholas Schiefer, Oliver Rausch, Robin Larson, Sam McCandlish, Sandipan Kundu, Saurav Kadavath, Shannon Yang, Thomas Henighan, Timothy Maxwell, Timothy Telleen-Lawton, Tristan Hume, Zac Hatfield-Dodds, Jared Kaplan, Jan Brauner, Samuel R. Bowman, and Ethan Perez. 2023. Measuring Faithfulness in Chain-of-Thought Reasoning. arXiv:2307.13702 [cs.AI]
- [58] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *arXiv preprint arXiv:2005.11401* (2020).
- [59] Haitao Li et al. 2024. LLMs-as-Judges: A Comprehensive Survey on LLM-based Evaluation Methods. *arXiv preprint arXiv:2412.05579* (2024).
- [60] Jia Li, Ge Li, Yongmin Li, and Zhi Jin. 2023. Structured Chain-of-Thought Prompting for Code Generation. *ACM Transactions on Software Engineering and Methodology* (2023). arXiv:2305.06599.
- [61] Wenhao Li et al. 2025. A Survey of Automatic Prompt Engineering: An Optimization Perspective. *arXiv preprint arXiv:2502.11560* (2025).
- [62] Zhen Li, Xiaohan Xu, Tao Shen, Can Xu, Jia-Chen Gu, Yuxuan Lai, Chongyang Tao, and Shuai Ma. 2024. Leveraging Large Language Models for NLG Evaluation: A Survey. arXiv:2401.07103 [cs.CL]
- [63] Bill Yuchen Lin et al. 2024. WildBench: Benchmarking LLMs with Challenging Tasks from Real Users in the Wild. *arXiv preprint arXiv:2406.04770* (2024).
- [64] Stephanie Lin, Jacob Hilton, and Owain Evans. 2021. TruthfulQA: Measuring How Models Mimic Human Falsehoods. *arXiv preprint arXiv:2109.07958* (2021).
- [65] Chengzhi Liu, Zhongxing Liang, Yifei Tian, Xueqian Sun, Zhuoxiao Tong, Yu Liu, Yi Liu, Hejia Cao, Pan Zhou, and Lichao Sun. 2025. More Thinking, Less Seeing? Assessing Amplified Hallucination in Multimodal Reasoning Models. arXiv:2505.21523 [cs.CV]
- [66] Jiawei Liu, Chunqiu Steven Xia, Yuyao Wang, and Lingming Zhang. 2023. Is Your Code Generated by ChatGPT Really Correct? Rigorous Evaluation of Large Language Models for Code Generation. *arXiv preprint arXiv:2305.01210*

(2023).

- [67] Weiwen Liu, Xu Huang, Xingshan Zeng, Xinlong Hao, Shuai Yu, Dexun Li, Shuai Wang, Weinan Gan, Zhengying Liu, Yuanqing Yu, Zezhong Wang, Yuxian Wang, Wu Ning, Yutai Hou, Bin Wang, Chuhan Wu, Xinzhi Wang, Yong Liu, Yasheng Wang, Duyu Tang, Dandan Liu, Huifeng Shang, Ruiming Wang, Ying Shen, and Qun Liu. 2024. ToolACE: Winning the Points of LLM Function Calling. *arXiv preprint arXiv:2409.00920* (2024).
- [68] Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, Minlie Huang, Yuxiao Dong, and Jie Tang. 2023. AgentBench: Evaluating LLMs as Agents. *arXiv:2308.03688* [cs.AI]
- [69] Yi Liu, Gelei Deng, Yuekang Li, Kailong Wang, Zihao Wang, Xiaofeng Wang, Tianwei Zhang, Yepang Liu, Haoyu Wang, Yan Zheng, and Yang Liu. 2023. Prompt Injection Attack against LLM-integrated Applications. *arXiv:2306.05499* [cs.CR]
- [70] Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment. *arXiv preprint arXiv:2303.16634* (2023).
- [71] Qing Lyu, Shreya Havaldar, Adam Stein, Li Zhang, Delip Rao, Eric Wong, Marianna Apidianaki, and Chris Callison-Burch. 2023. Faithful Chain-of-Thought Reasoning. *arXiv preprint arXiv:2301.13379* (2023).
- [72] Chang Ma, Junlei Zhang, Zhihao Zhu, Cheng Yang, Yujiu Yang, Yaohui Jin, Zhenzhong Lan, Lingpeng Kong, and Junxian He. 2024. AgentBoard: An Analytical Evaluation Board of Multi-Turn LLM Agents. *arXiv:2401.13178* [cs.CL]
- [73] Saumya Malik, Valentina Pyatkin, Sander Land, Jacob Morrison, Noah A. Smith, Hannaneh Hajishirzi, and Nathan Lambert. 2025. RewardBench 2: Advancing Reward Model Evaluation. *arXiv:2506.01937* [cs.CL]
- [74] Narek Maloyan, Bulat Nutfullin, and Eugene Ilyushin. 2025. Investigating the Vulnerability of LLM-as-a-Judge Architectures to Prompt-Injection Attacks. *arXiv:2505.13348* [cs.CR]
- [75] Timothy R. McIntosh, Teo Susnjak, Tong Liu, Paul Watters, and Malka N. Halgamuge. 2024. Inadequacies of Large Language Model Benchmarks in the Era of Generative AI. *IEEE Transactions on Artificial Intelligence* (2024). doi:10.1109/TAI.2024.3394596
- [76] Sewon Min, Kalpesh Krishna, Xinxin Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation. *arXiv preprint arXiv:2305.14251* (2023).
- [77] Iman Mirzadeh, Keivan Alizadeh, Hooman Shahrokhi, Oncl Tuzel, Samy Bengio, and Mehrdad Farajtabar. 2024. GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models. *arXiv:2410.05229* [cs.LG]
- [78] Moran Mizrahi, Guy Kaplan, Dan Malkin, Rotem Dror, Roi Schwartz, and Gabriel Stanovsky. 2024. State of What Art? A Call for Multi-Prompt LLM Evaluation. *Transactions of the Association for Computational Linguistics* 12 (2024), 933–949. doi:10.1162/tacl_a_00681
- [79] Mahmoud Mohammadi, Yipeng Li, Jane Lo, and Wendy Yip. 2025. Evaluation and Benchmarking of LLM Agents: A Survey. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '25)*. *arXiv:2507.21504*.
- [80] Shiwen Ni et al. 2025. A Survey on Large Language Model Benchmarks. *arXiv preprint arXiv:2508.15361* (2025).
- [81] Harsha Nori, Nicholas King, Scott Mayer McKinney, Dean Carignan, and Eric Horvitz. 2023. Capabilities of GPT-4 on Medical Challenge Problems. *arXiv preprint arXiv:2303.13375* (2023).
- [82] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. 2022. Training Language Models to Follow Instructions with Human Feedback. *arXiv preprint arXiv:2203.02155* (2022).
- [83] Arjun Panickssery, Samuel R. Bowman, and Shi Feng. 2024. Judging the Judges: Evaluating Alignment and Vulnerabilities in LLMs-as-Judges. *arXiv preprint arXiv:2406.12624* (2024).
- [84] Felipe Maia Polo, Ronald Xu, Lucas Weber, Mírian Silva, Onkar Bhardwaj, Leshem Choshen, Allysson Flavio Melo de Oliveira, Yuekai Sun, and Mikhail Yurochkin. 2024. Efficient Multi-Prompt Evaluation of LLMs. *arXiv:2405.17202* [cs.CL]
- [85] Reid Pryzant, Dan Iter, Jerry Li, Yin Tat Lee, Chenguang Zhu, and Michael Gao. 2023. Automatic Prompt Optimization with “Gradient Descent” and Beam Search. *arXiv preprint arXiv:2305.03495* (2023).
- [86] Chengwei Qin, Aston Zhang, Zhuosheng Chen, Jiaao Chen, Michihiro Yasunaga, and Diyi Yang. 2024. Is ChatGPT a Good Sentiment Reasoner? A Preliminary Study. *arXiv:2401.03601* [cs.CL]
- [87] Mingqi Qin, Yan Shen, Jihua Yan, Jiahao Ye, and Xiaojun Hu. 2024. LLM-based NLG Evaluation: Current Status and Challenges. *arXiv preprint arXiv:2402.01383* (2024).
- [88] Ori Ram, Yoav Levine, Itay Dalmedigos, Dor Muhlgay, Amnon Shashua, Kevin Leyton-Brown, and Yoav Shoham. 2023. In-Context Retrieval-Augmented Language Models. *Transactions of the Association for Computational Linguistics* 11 (2023), 1316–1331.

- [89] Kiran Ramnath et al. 2025. A Systematic Survey of Automatic Prompt Optimization Techniques. *arXiv preprint arXiv:2502.16923* (2025).
- [90] Amirhossein Razavi, M. Soltangheis, Negar Arabzadeh, S. Salamat, M. Zihayat, and E. Bagheri. 2025. Benchmarking Prompt Sensitivity in Large Language Models. In *Advances in Information Retrieval (ECIR 2025) (Lecture Notes in Computer Science, Vol. 15574)*. Springer. *arXiv:2502.06065*.
- [91] Baptiste Rozière, Jonas Gehring, Fabian Gloeckle, Sten Sootla, Itai Gat, Xiaoqing Ellen Tan, Yossi Adi, Jingyu Liu, Tal Remez, Jérémy Rapin, et al. 2023. Code Llama: Open Foundation Models for Code. *arXiv preprint arXiv:2308.12950* (2023).
- [92] Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. 2023. ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems. *arXiv preprint arXiv:2311.09476* (2023).
- [93] Pranab Sahoo, Ayush Kumar Singh, Sriparna Saha, Vinija Jain, Samrat Mondal, and Aman Chadha. 2024. A Systematic Survey of Prompt Engineering in Large Language Models: Techniques and Applications. *arXiv preprint arXiv:2402.07927* (2024).
- [94] Oscar Sainz, Jon Ander Campos, Iker Garcia-Ferrero, Julen Etxaniz, Oier Lopez de Lacalle, and Eneko Agirre. 2023. NLP Evaluation in trouble: On the Need to Measure LLM Data Contamination for each Benchmark. *arXiv:2310.18018* [cs.CL]
- [95] Abel Salinas and Fred Morstatter. 2024. The Butterfly Effect of Altering Prompts: How Small Changes and Jailbreaks Affect Large Language Model Performance. *arXiv preprint arXiv:2401.03729* (2024).
- [96] David Salinas and Fabian Gama. 2024. Does Prompt Formatting Have Any Impact on LLM Performance? *arXiv preprint arXiv:2411.10541* (2024).
- [97] Yuichi Sasaki, Hironori Washizaki, Jiawei Li, David Sander, Nobukazu Yoshioka, and Yoshiaki Fukazawa. 2025. Landscape and Taxonomy of Prompt Engineering Patterns in Software Engineering. *IT Professional* 27, 1 (2025), 41–50. DOI: 10.1109/MITP.2024.3525458.
- [98] Sander Schulhoff, Michael Ilie, Nishant Balepur, Konstantine Kahadze, Amanda Liu, Chenglei Si, Yinheng Li, Aayush Gupta, HyoJung Han, Sevien Schulhoff, et al. 2024. The Prompt Report: A Systematic Survey of Prompting Techniques. *arXiv preprint arXiv:2406.06608* (2024).
- [99] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2023. Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design or: How I Learned to Start Worrying about Prompt Formatting. *arXiv preprint arXiv:2310.11324* (2023).
- [100] Mrinank Sharma, Meg Tong, Jesse Mu, Jerry Wei, Jorrit Kruthoff, Scott Goodfriend, Euan Ong, Alwin Peng, Raj Agarwal, Cem Anil, et al. 2025. Constitutional Classifiers: Defending against Universal Jailbreaks across Thousands of Hours of Red Teaming. *arXiv:2501.18837* [cs.CL]
- [101] Freda Shi, Mirac Suzgun, Markus Freitag, Xuezhi Wang, Suraj Srivats, Soroush Vosoughi, Hyung Won Chung, Yi Tay, Sebastian Ruder, Denny Zhou, Dipanjan Das, and Jason Wei. 2022. Language Models are Multilingual Chain-of-Thought Reasoners. *arXiv preprint arXiv:2210.03057* (2022).
- [102] Taylor Shin, Yasaman Razeghi, Robert L. Logan IV, Eric Wallace, and Sameer Singh. 2020. AutoPrompt: Eliciting Knowledge from Language Models with Automatically Generated Prompts. *arXiv preprint arXiv:2010.15980* (2020).
- [103] Sonish Sivarajkumar, Mark Kelley, Alyssa Samolyk-Mazzanti, Shyam Visweswaran, and Yanshan Wang. 2024. An Empirical Evaluation of Prompting Strategies for Large Language Models in Zero-Shot Clinical Natural Language Processing. *JMIR Medical Informatics* 12 (2024). doi:10.2196/55318
- [104] Guijin Son, Dongkeun Yoon, Juyoung Suk, Javier Aula-Blasco, Mano Aslan, Vu Trong Kim, Shayekh Bin Islam, Jaume Prats-Cristia, Lucia Tormo Hurtado, Sebastian Gehrmann, Anna Heintz-Buschart, Hyeon-Jin Cho, Jiwoo Kim, Sangha Choi, and Hyunwoo Lim. 2024. MM-Eval: A Multilingual Meta-Evaluation Benchmark for LLM-as-a-Judge and Reward Models. *arXiv:2410.17578* [cs.CL]
- [105] Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, Abubakar Abid, Adam Fisch, et al. 2022. Beyond the Imitation Game: Quantifying and Extrapolating the Capabilities of Language Models. *arXiv preprint arXiv:2206.04615* (2022).
- [106] Mark Steyvers, Heliodoro Tejeda, Aakriti Kumar, Catarina Belem, Sheer Karny, Xinyue Hu, Lukas W. Mayer, and Padhraic Smyth. 2024. What Large Language Models Know and What People Think They Know. *Nature Machine Intelligence* (2024). doi:10.1038/s42256-024-00976-7
- [107] James W. A. Strachan, Dalila Albergio, Giulia Borghini, Oriana Pansardi, Eugenio Scaliti, Saurabh Gupta, Krati Saxena, Alessandro Rufo, Stefano Panzeri, Guido Manzi, Michael S. A. Graziano, and Cristina Becchio. 2024. Testing Theory of Mind in Large Language Models and Humans. *Nature Human Behaviour* (2024). doi:10.1038/s41562-024-01882-z
- [108] Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V. Le, Ed H. Chi, Denny Zhou, and Jason Wei. 2022. Challenging BIG-Bench Tasks and Whether Chain-of-Thought Can Solve Them. *arXiv preprint arXiv:2210.09261* (2022).
- [109] Sijun Tan et al. 2024. JudgeBench: A Benchmark for Evaluating LLM-based Judges. *arXiv preprint arXiv:2410.12784* (2024).

- [110] Haoye Tian, Chong Wang, Boyang Yang, Lyuye Zhang, and Yang Liu. 2025. A Taxonomy of Prompt Defects in LLM Systems. *arXiv preprint arXiv:2509.14404* (2025).
- [111] Cathrin Tony, Markus Mutas, Nicolas E. Diaz Ferreyra, and Riccardo Scandariato. 2024. Prompting Techniques for Secure Code Generation: A Systematic Investigation. *ACM Transactions on Software Engineering and Methodology* (2024). [arXiv:2407.07064](#).
- [112] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023. LLaMA: Open and Efficient Foundation Language Models. *arXiv preprint arXiv:2302.13971* (2023).
- [113] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shrutti Bhosale, et al. 2023. Llama 2: Open Foundation and Fine-Tuned Chat Models. *arXiv preprint arXiv:2307.09288* (2023).
- [114] Shubham Vatsal and Harsh Dubey. 2024. A Survey of Prompt Engineering Methods in Large Language Models for Different NLP Tasks. *arXiv preprint arXiv:2407.12994* (2024).
- [115] Shubham Vatsal and Harsh Singh. 2025. Multilingual Prompt Engineering in Large Language Models: A Survey Across NLP Tasks. [arXiv:2505.11665](#) [cs.CL]
- [116] Pat Verga, Sebastian Hofstatter, Sophia Holenstein, Jordan Boyd-Graber, Carrie Lawrence, Jennifer Ni, Rodrigo Nogueira, Shyam Upadhyay, Yura Mansour, and Slav Petrov. 2024. Replacing Judges with Juries: Evaluating LLM Generations with a Panel of Diverse Models. [arXiv:2404.18796](#) [cs.CL]
- [117] Alex Wang, Yada Pruksachatkun, Nikita Nangia, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel Bowman. 2019. SuperGLUE: A Stickier Benchmark for General-Purpose Language Understanding Systems. *arXiv preprint arXiv:1905.00537* (2019).
- [118] Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. 2018. GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding. *arXiv preprint arXiv:1804.07461* (2018).
- [119] Lei Wang, Wanyu Xu, Yihuai Lan, Zhiqiang Hu, Yunshi Lan, Roy Ka-Wei Lee, and Ee-Peng Lim. 2023. Plan-and-Solve Prompting: Improving Zero-Shot Chain-of-Thought Reasoning by Large Language Models. *arXiv preprint arXiv:2305.04091* (2023).
- [120] Siyuan Wang, Zhuohan Sun, Zhichao Li, Zhenyu Gao, Chenxi Wang, Rui Chen, Sichu Li, Lifan He, Hao Xu, et al. 2024. Benchmark Self-Evolving: A Multi-Agent Framework for Dynamic LLM Evaluation. [arXiv:2402.11443](#) [cs.CL]
- [121] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. *arXiv preprint arXiv:2203.11171* (2023).
- [122] Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhramil Chandra, Shengbang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, et al. 2024. MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark. *arXiv preprint arXiv:2406.01574* (2024).
- [123] Taku Wataoka, Tatsushi Oba, Yusuke Iwasawa, and Yutaka Matsuo. 2024. Self-Preference Bias in LLM-as-a-Judge. [arXiv:2410.21819](#) [cs.CL]
- [124] Alexander Wei, Nika Haghtalab, and Jacob Steinhardt. 2023. Jailbroken: How Does LLM Safety Training Fail? *arXiv preprint arXiv:2307.02483* (2023).
- [125] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. *arXiv preprint arXiv:2201.11903* (2022).
- [126] Jianguo Wen et al. 2025. A Comprehensive Taxonomy of Prompt Engineering Techniques for Large Language Models. *Frontiers of Computer Science* (2025).
- [127] Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C. Schmidt. 2023. A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT. *arXiv preprint arXiv:2302.11382* (2023).
- [128] Guangzhi Xiong, Qiao Jin, Zhiyong Lu, and Aidong Zhang. 2024. Benchmarking Retrieval-Augmented Generation for Medicine. *arXiv preprint arXiv:2402.13178* (2024).
- [129] Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. 2023. Can LLMs Express Their Uncertainty? An Empirical Evaluation of Confidence Elicitation in LLMs. *arXiv preprint arXiv:2306.13063* (2023).
- [130] Wenda Xu, Danqing Wang, Liangming Pan, Zhenqiao Song, Philippe Laban, William Yang Wang, et al. 2023. INSTRUCTSCORE: Towards Explainable Text Generation Evaluation with Automatic Feedback. *arXiv preprint arXiv:2305.14282* (2023).
- [131] Hui Yang et al. 2024. An Empirical Study of LLM-as-a-Judge for LLM Evaluation: Fine-tuned Judge Models Are Task-Specific Classifiers. *arXiv preprint arXiv:2403.02839* (2024).
- [132] Shuo Yang, Wei-Lin Chiang, Lianmin Zheng, Joseph E. Gonzalez, and Ion Stoica. 2023. Rethinking Benchmark and Contamination for Language Models with Rephrased Samples. [arXiv:2311.04850](#) [cs.CL]

- [133] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. 2024. τ -bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains. *arXiv preprint arXiv:2406.12045* (2024).
- [134] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izheng Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. 2023. Tree of Thoughts: Deliberate Problem Solving with Large Language Models. *arXiv preprint arXiv:2305.10601* (2023).
- [135] Guiming Hardy Ye et al. 2024. Humans or LLMs as the Judge? A Study on Judgement Bias. *arXiv preprint arXiv:2402.10669* (2024).
- [136] Jiayi Ye, Yanbo Wang, Yue Huang, Dongping Chen, Qihui Zhang, Nuno Moniz, Tian Gao, Werner Geyer, Chao Huang, Pin-Yu Chen, Nitesh V. Zou, and Xiangliang Liu. 2024. Justice or Prejudice? Quantifying Biases in LLM-as-a-Judge. *arXiv:2410.02736* [cs.CL]
- [137] Howard Yen, Tianyu Gao, Minmin Hou, Ke Ding, Daniel Fleischer, Peter Izsak, Moshe Wasserblat, and Danqi Chen. 2024. HELMET: How to Evaluate Long-Context Language Models Effectively and Thoroughly. *arXiv:2410.02694* [cs.CL]
- [138] Siboyi, Yule Liu, Zhen Sun, Tianshuo Cong, Xinlei He, Jiaxing Song, Ke Xu, and Qi Li. 2024. Jailbreak Attacks and Defenses Against Large Language Models: A Survey. *arXiv preprint arXiv:2407.04295* (2024).
- [139] Fangyi Yu, Nabeel Seedat, Drahomira Hermannova, Frank Schilder, and Jonathan Richard Schwarz. 2025. Beyond Pointwise Scores: Decomposed Criteria-Based Evaluation of LLM Responses. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*. Association for Computational Linguistics, 1931–1954.
- [140] Hao Yu, Aoran Gan, Kai Zhang, Shiwei Tong, Qi Liu, and Zhaofeng Liu. 2024. Evaluation of Retrieval-Augmented Generation: A Survey. *arXiv:2405.07437* [cs.CL]
- [141] Jiahao Yu, Xingwei Lin, Zheng Yu, and Xinyu Xing. 2023. GPTFuzzer: Red Teaming Large Language Models with Auto-Generated Jailbreak Prompts. *arXiv preprint arXiv:2309.10253* (2023).
- [142] Zhiyuan Zeng, Jiatong Yu, Tianyu Gao, Yu Meng, Tanya Goyal, and Danqi Chen. 2023. Evaluating Large Language Models at Evaluating Instruction Following. *arXiv:2310.07641* [cs.CL]
- [143] Haiyan Zhao, Hanjie Chen, Fan Yang, Ninghao Liu, Huiqi Deng, Hengyi Cai, Shuaiqiang Wang, Dawei Yin, and Mengnan Du. 2024. Explainability for Large Language Models: A Survey. *ACM Transactions on Intelligent Systems and Technology* 15, 2 (2024). *arXiv:2309.01029*.
- [144] Penghao Zhao, Hailin Zhang, Qinhan Yu, Zhengren Wang, Yunteng Geng, Fangcheng Fu, Ling Yang, Wentao Zhang, and Bin Cui. 2024. Retrieval-Augmented Generation for AI-Generated Content: A Survey. *arXiv preprint arXiv:2402.19473* (2024).
- [145] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, et al. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *arXiv preprint arXiv:2306.05685* (2023).
- [146] Ming Zhong, Yang Liu, Da Chen, Danqing Zhu, Jiawei Gong, Chuanqi Zhang, Chongyang Duan, Xipeng Qiu, Qi Zhang, Luke Zettlemoyer, and Pengcheng Jiang. 2022. Towards a Unified Multi-Dimensional Evaluator for Text Generation. *arXiv preprint arXiv:2210.07197* (2022).
- [147] Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Olivier Bousquet, Quoc V. Le, and Ed H. Chi. 2022. Least-to-Most Prompting Enables Complex Reasoning in Large Language Models. *arXiv preprint arXiv:2205.10625* (2022).
- [148] Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny Zhou, and Le Hou. 2023. Instruction-Following Evaluation for Large Language Models. *arXiv:2311.07911* [cs.CL]
- [149] Yongchao Zhou, Andrei Ioan Muresanu, Ziwen Han, Keiran Paster, Silviu Pitis, Harris Chan, and Jimmy Ba. 2022. Large Language Models Are Human-Level Prompt Engineers. *arXiv preprint arXiv:2211.01910* (2022).
- [150] Zhengping Zhou et al. 2026. Benchmark²: Systematic Evaluation of LLM Benchmarks. *arXiv preprint arXiv:2601.03986* (2026).
- [151] Kaijie Zhu, Jindong Wang, Jiaheng Zhou, Zichen Wang, Hao Chen, Yidong Wang, Linyi Yang, Wei Ye, Neil Zhenqiang Gong, Yue Zhang, and Xing Xie. 2023. PromptBench: Towards Evaluating the Robustness of Large Language Models on Adversarial Prompts. *arXiv preprint arXiv:2306.04528* (2023).
- [152] Lianghui Zhu, Xinggang Wang, and Xinlong Wang. 2023. JudgeLM: Fine-tuned Large Language Models are Scalable Judges. *arXiv:2310.17631* [cs.CL]